



UNIVERSITY OF THE PUNJAB

Seventh Semester 2017
Examination: B.S. 4 Years Programme

Roll No.

PAPER: Time Series Analysis-I
Course Code: STAT-411

TIME ALLOWED: 2 hrs. & 30 mins.
MAX. MARKS: 50

Attempt this Paper on Separate Answer Sheet provided.

SUBJECTIVE TYPE

Q2.

(4 x 5=20)

- Define time series. Differentiate between discrete and continuous time series.
- Define strict and weak stationarity.
- Show that random walk is a non-stationary process.
- Show that $Y_t = Y_{t-1} - \lambda Y_{t-2} + \varepsilon_t$ is stationary when $-1 < \lambda < 0$.

Q3. The following table gives sample time series:

(3 + 7=10)

t	1	2	3	4	5	6	7	8	9	10
Y_t	5	7	7	6	8	9	9	12	13	14

- Plot the time series. Do you think the plot exhibits stationarity? Provide reasoning.
- Take first order differences i.e. $W_t = Y_t - Y_{t-1}$. Calculate autocorrelation at lag 1.

Q4. Show that for the ARMA(1,1) process $Y_t = \phi Y_{t-1} + Z_t + \theta Z_{t-1}$

(10)

$$\rho_k = \begin{cases} (1 + \phi\theta)(\phi + \theta) / (1 + \theta^2 + \phi\theta) & k = 1 \\ \phi\rho_{k-1} & k = 2, 3, \dots \end{cases}$$

Q5. Show that the autocorrelation function of $Y_t = \frac{1}{3}Y_{t-1} + \frac{2}{9}Y_{t-2} + Z_t$ is given by

(10)

$$\rho_k = \frac{16}{21} \left(\frac{2}{3}\right)^k + \frac{5}{21} \left(-\frac{1}{3}\right)^k, \quad k = 0, 1, 2, 3, \dots$$

where $\{Z_t\}$ is a purely random process having zero mean and finite variance.





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Attempt this Paper on Separate Answer Sheet provided.

Note: Attempt all questions.

Question No.2. Describe the following.

(5 × 4 = 20)

- Derive the condition of stationarity for an AR(1) process.
- Define autocorrelation and write down the properties of autocorrelation coefficient.
- Find autocorrelation function of MA(2) process given by $Y_t = Z_t + 0.7Z_{t-1} - 0.2Z_{t-2}$.
- Show that for an ARMA(p, q) process, $\varphi(B) \rho_k = 0$; for $k \geq q + 1$, where q is the order of MA component.

Question No. 3. a) The first ten sample autocorrelation coefficients of 400 random numbers are given below;

(05)

K	1	2	3	4	5	6	7	8	9	10
r_k	0.02	0.05	-0.09	0.08	-0.02	0.0	0.12	0.06	0.02	-0.08

Is there any evidence of non-randomness? Give reasons for your answer.

b) Define Seasonal ARIMA model, write down the model for SARIMA (1,0,1) × (0,0,2)₆

(05)

Question No. 4. Show that the ACF of the following AR(2) stationary process

(10)

$$X_t = X_{t-1} - \frac{1}{2}X_{t-2} + Z_t$$

is given by,

$$\rho_K = \left(\frac{1}{\sqrt{2}}\right)^{|K|} \left| \cos \frac{|K\pi|}{4} + \frac{1}{3} \sin \frac{|K\pi|}{4} \right| ; \text{ for } K = 0, \pm 1, \pm 2, \pm 3, \dots$$

Question No. 5. A time series model is given below ,

(10)

$$X_t = 0.6X_{t-1} + Z_t - 0.7Z_{t-1}$$

- Identify the model.
- Check stationarity and invertibility of the model.
- Obtain first three ψ - weights of the model when expressed as an MA model.
- Obtain first three π - weights of the model when expressed as an AR model.



UNIVERSITY OF THE PUNJAB

Seventh Semester – 2019

Examination: B.S. 4 Years Program

Roll No.

PAPER: Time Series Analysis-I

Course Code: STAT-411 Part – II

MAX. TIME: 2 Hrs. 45 Min.

MAX. MARKS: 50

ATTEMPT THIS (SUBJECTIVE) ON THE SEPARATE ANSWER SHEET PROVIDED

Q.2. Answer these short questions.

(4x5=20)

- a) Differentiate between weak and strict stationarity.
- b) Consider an AR(p) process, $X_t = \varphi_1 X_{t-1} + \varphi_2 X_{t-2} + \dots + \varphi_k X_{t-k} + Z_t$,
Show that,
$$\sigma_X^2 = \frac{\sigma_Z^2}{1 - \varphi_1 \rho_1 - \varphi_2 \rho_2 - \dots - \varphi_k \rho_k}$$
- c) Define random walk process; show that its mean and variance are time dependent.
- d) Define Partial autocorrelation. Given the autocorrelations at lag 1, 2 & 3 of a time series as $-0.55, 0.25$ & -0.014 respectively. Calculate partial autocorrelation at lag-3.

Question No. 3. A time series model is given below,

(12)

$$(1 - B)(1 - 0.2B)X_t = (1 - 0.5B)Z_t$$

- a) Identify the model of X_t .
- b) Check stationarity and invertibility of the model.
- c) Obtain first three ψ - weights of the model when expressed as an MA model.
- d) Obtain first three π - weights of the model when expressed as an AR model.

Question No. 4. Show that AR(2) process ,

(08)

$X_t = \varphi_1 X_{t-1} + \varphi_2 X_{t-2} + Z_t$, will be stationary if;

$$\varphi_1 + \varphi_2 < 1$$

$$\varphi_2 - \varphi_1 < 1$$

$$\varphi_2 > -1$$

Question No. 5. Consider the infinite order MA process $\{X_t\}$, defined by

(10)

$X_t = Z_t + C(Z_{t-1} + Z_{t-2} + \dots)$, where C is a constant. Show that process is non-stationary. Also show that the series of first differences $\{Y_t\}$, defined by $Y_t = X_t - X_{t-1}$, is a first order MA process and is stationary. Also find the ACF of $\{Y_t\}$.



UNIVERSITY OF THE PUNJAB

B.S. 4 Years Program : Seventh Semester – Fall 2021

Paper: Time Series Analysis-I

Course Code: STAT-411

Roll No.

Time: 3 Hrs. Marks: 60

Q.1. Answer the following short questions:

(6x5=30)

- Discuss the duality between moving average and auto-regressive process.
- Write a short note on stationarity of time series.
- Define stochastic process.
- Discuss about the invertibility of MA(q) process.
- Write down the objectives of time series analysis.
- Define Autocorrelation and Correlogram.

Answer the following questions.

(3x10=30)

Q.2. Given the following time series, compute the first two partial autocorrelations.

Year	2000	2001	2002	2003	2004	2005	2006	2007	2008	2009
X_t	13	8	15	4	4	12	11	7	14	12

Q.3. Derive the properties of AR(p) process.

Q.4. Given the following MA(1) processes

$$\text{Model-1 : } X_t = Z_t + \beta Z_{t-1}$$

$$\text{Model-2 : } X_t = Z_t + \frac{1}{\beta} Z_{t-1}$$

Where $\{Z_t\}$ is a purely random process with zero mean and finite variance σ^2

Find the autocorrelation functions of Model-1 and Model-2. Also comment on the invertibility condition of MA process by discussing the autocorrelation functions of both models.



UNIVERSITY OF THE PUNJAB

B.S. 4 Years Program / Seventh Semester – Fall 2024

Paper: Time Series Analysis-I

Course Code: STAT-411

Roll No.

Marks: 60

THE ANSWERS MUST BE ATTEMPTED ON THE ANSWER SHEET PROVIDED

Q.1. Answer the following short questions: (5x6=30)

- Discuss the duality between moving average and auto-regressive process.
- Write a note on stationary time series? Why it is important to transform a non-stationary time series into stationary time series?
- Define mixed ARMA(p,q) process also comment on the stationarity and invertibility of ARMA (p,q) process.
- Write briefly about secular trend and irregular variations in time series.
- Discuss autocorrelation and partial autocorrelation.

Answer the following questions. (3x10=30)

Q.2. The following table shows the number of bags (in hundreds) of fertilizer sold by a certain dealer. Compute the indices of seasonal variation by the ratio-to-trend method by fitting a least square straight line trend to the observed time series data.

Year	Quarters			
	I	II	III	IV
2001	72	98	79	106
2002	79	122	101	143
2003	94	141	128	160
2004	125	143	135	187

Q.3. Prove that for a stationary time series $\text{Corr}(X_t, X_{t-1}) = \text{Corr}(X_t, X_{t+1})$

Q.4. Find auto correlation for AR(1) process given below at lag 1,2 and 3.

$$X_t = Z_t + 0.9 X_{t-1}$$



UNIVERSITY OF THE PUNJAB
B.S. 4 Years Program : Seventh Semester – 2021

Paper: Time Series Analysis-I
Course Code: STAT-411

Part – II

Time: 2 Hrs. 45 Min. Marks

Roll No.

ATTEMPT THIS (SUBJECTIVE) ON THE SEPARATE ANSWER SHEET PROVIDED

Q.2. Give short answers of the following: (4x5=20)

- Define components of time series with the help of examples.
- Write a brief note on special features of time series data.
- Describe the Principle of Parsimony.
- Check the invertibility and Stationarity of the process given below:

$$X_t = 0.5 X_{t-1} + Z_t + 0.3 Z_{t-1}$$

Solve the following questions. (3x10=30)

Q3. 16 successive observations on a stationary time series are as follows:

t	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16
X_t	1.6	0.8	1.2	0.5	0.9	1.1	1.1	0.6	1.5	0.8	0.9	1.2	0.5	1.3	0.8	1.2

Find Autocorrelation for $k = 1, 2, 3$.

Q.4. Determine a trend line by a simple moving average of 5 years from the following data:

Year	2001	2002	2003	2004	2005	2006	2007	2008	2009	2010
Value	102	108	130	140	158	180	196	210	220	230

Q.5. (a): For AR(P) process, show that:

$$\rho_k = \varphi_1 \rho_{k-1} + \varphi_2 \rho_{k-2} + \dots + \varphi_p \rho_{k-p} \quad \text{for } k > 0$$

(b): For AR(3) process, show that

$$\rho_1 = \frac{\varphi_1 + \varphi_2 \varphi_3}{1 - \varphi_2 - \varphi_3 - \varphi_1 \varphi_3}$$

by using recursive rule.



UNIVERSITY OF THE PUNJAB

B.S. 4 Years Program / Seventh Semester – Fall 2023

Paper: Time Series Analysis-I

Course Code: STAT-411

(Common with BS-AD 7th)

Roll No.

Time: 3 Hrs. Marks: 60

THE ANSWERS MUST BE ATTEMPTED ON THE ANSWER SHEET PROVIDED

Q.1. Answer the following short questions:

(6×5=30)

- Write a note on stabilizing the mean and variance in case of non-stationary time series.
- Write a short note on stationarity of a time series.
- Differentiate between deterministic and stochastic time series.
- Define secular trend and seasonal variation.
- Describe the basic aims of analyzing a time series?
- Write a short note on serial autocorrelation.

Q.2. Answer the following questions.

(3×10=30)

1. The following are the annual profits in thousands of rupees in a certain business:

Year	2017	2018	2019	2020	2021	2022	2023
Profit	88	101	108	91	113	120	132

- For AR(1) process, find autocorrelation function at lag k.
- Find the ψ weights and π weights for the ARMA (1,1) process given by

$$X_t = 0.5 X_{t-1} + Z_t - 0.3 Z_{t-1}$$

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